

Python

Home Assignments

PRAVEEN NAIR

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Part 1

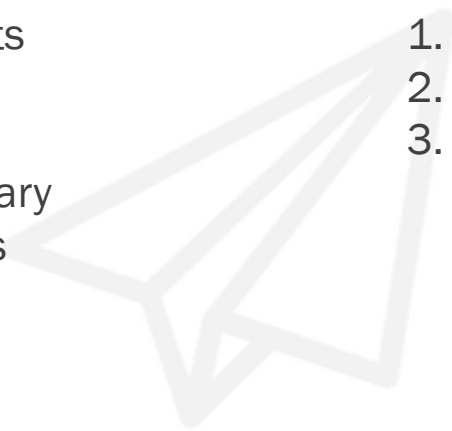
1. Variables and Constants
2. If-elif-else
3. While loop and for loop
4. List, Tuple, Set, Dictionary
5. Functions and Modules

Part 2 (In-built Modules)

1. random
2. os

Part 3 (External Modules)

1. Numpy
2. Pandas
3. Matplotlib



Assignment: Student Information and Marks

Write a Python program that accepts the following details from the user using **Scanner**:

1. Student name
2. Student age
3. Student roll number
4. Marks in Java
5. Marks in Python
6. Marks in Database

Calculate and display:

1. Student details
2. Total marks
3. Average marks

No if or loops or functions



Assignment: Employee Salary Calculator

Write a Python program that accepts the following details from the user:

1. Employee name
2. Employee ID
3. Basic salary
4. HRA
5. DA

Calculate and display:

1. Employee details
2. Gross salary = Basic Salary + HRA + DA
3. Annual salary = Gross Salary $\times$ 12

No if or loops or functions

Assignment: Shopping Bill Calculator

Write a Python program that accepts:

1. Customer name
2. Product name
3. Product price
4. Quantity

Calculate and display:

1. Total price
2. Discount of 10%
3. Final bill amount after discount

No if or loops or functions



Student Result & Grade Calculator

Create a Python program that accepts a student's marks in **5 subjects**, calculates the total and percentage, and determines:

Pass or Fail

Grade

Division

Scholarship eligibility

Note: Do not use loops, arrays, methods, classes beyond the main class, or switch

Use if statement

Requirements

The student enters:

Student name

Java marks

Python marks

Database marks

Web Technology marks

Aptitude marks

Each subject is out of **100**.



Condition 1 - Validate marks

If any mark is:

Less than 0 → Invalid marks

Greater than 100 → Invalid marks



Condition 2 - Pass/Fail

A student passes only if **each subject** is at least 40.

Otherwise:

Result: FAIL

Reason: One or more subjects below 40



Condition 3 - Grade

Percentage	Grade
90-100	A+
80-89	A
70-79	B
60-69	C
50-59	D
40-49	E

Condition 4 - Division

Percentage	Division
75+	Distinction
60-74	First Class
50-59	Second Class
40-49	Third Class

Condition 5 - Scholarship

Display:

Scholarship: Eligible

only when:

percentage ≥ 85 AND every subject ≥ 75

Otherwise:

Scholarship: Not Eligible



Sample Output

===== STUDENT RESULT SYSTEM =====

Enter student name: Rahul

Enter Java marks: 85

Enter Python marks: 78

Enter Database marks: 92

Enter Web Technology marks: 88

Enter Aptitude marks: 81

===== RESULT =====

Student Name : Rahul

Total : 424 / 500

Percentage : 84.8%

Result : PASS

Grade : A

Division : First Class

Scholarship : Not Eligible



Bank Management System

Create a **console-based Bank Management System** in Python. The program should allow a user to deposit money, withdraw money, check the current balance, and exit the application.

Initial Balance

The account should start with an initial balance of ₹5,000.

Main Menu

When the program starts, display the following menu:

===== BANK MANAGEMENT SYSTEM =====

1. Deposit
2. Withdraw
3. View Balance
4. Exit

The menu should continue to appear until the user selects **Exit**.

Use if and loop if necessary

Expected Output

===== BANK MANAGEMENT SYSTEM =====

1. Deposit
2. Withdraw
3. View Balance
4. Exit

Enter your choice: 1
Enter deposit amount: 2000
Amount deposited successfully.
New Balance: 7000.0

===== BANK MANAGEMENT SYSTEM =====

1. Deposit
2. Withdraw
3. View Balance
4. Exit

Enter your choice: 4
Thank you for using the Bank Management System.



Employee Payroll System

The program should display:

===== EMPLOYEE PAYROLL SYSTEM =====

1. Enter Employee Detailsg
2. Calculate Gross Salary
3. Calculate Tax
4. Display Salary Slip
5. Exit

use if, loop and functions

1. Enter Employee Details

Ask the user to enter:

Employee name

Basic salary

HRA

Other allowance

Functions Students Should Practice

enterEmployeeDetails()

calculateGrossSalary()

calculateTax(grossSalary())

displaySalarySlip()

Expected Output

===== EMPLOYEE PAYROLL SYSTEM =====

1. Enter Employee Details
2. Calculate Gross Salary
3. Calculate Tax
4. Display Salary Slip
5. Exit

Enter your choice: 4

===== SALARY SLIP =====

Employee Name : John
Basic Salary : 10000.0
HRA : 2000.0
Allowance : 3000.0

Gross Salary : 15000.0
Tax : 0.0
Net Salary : 15000.0

=====

Student Record File Manager

Create a **Student Record File Manager** that stores student information in a text file and allows the user to:

===== Student Record Manager =====

1. Add Student
2. View All Students
3. Search Student
4. Exit

Use if, loop, function and os module

Use a file named:
students.txt

Each student record should be stored in the following format:

101,John,Java,85
102,Ravi,Python,78
103,Anita,React,92

Format:
Roll No,Name,Course,Marks

Thank You

- PRAVEEN NAIR